



Abdalrhman's Incident handler's journal

Date: Monday, March 16, 2026 1:23 PM	Entry: 1
Description	A ransomware incident caused by a phishing email
Tool(s) used	List any cybersecurity tools that were used.
The 5 W's	Capture the 5 W's of an incident. • Who caused the incident? An organized group of unethical hackers. • What happened? The hacker group sent a phishing email that contained a malicious attachment. The malicious attachment was downloaded and attackers gained access to the system. The attackers then deployed a ransomware attack encrypting the organization's computer files. A note was left by an organized hacker group to ransom the files. • When did the incident occur? Tuesday at 9:00 a.m • Where did the incident happen? Organization computer files • Why did the incident happen? A malicious attachment was downloaded. No employee education in place.
Additional notes	Include any additional thoughts, questions, or findings. Is there any security education in place?.

Date: Wednesday, March 25, 2026 3:06 PM	Entry: 2
Description	Investigating a phishing attack
Tool(s) used	VirusTotal VirusTotal has been used to analysis the malicious attachment. The tool has provided valuable information. First, it confirmed the malicious intent. Second, it linked it to an APT group. Furthermore, it gave meticulous details of the innerworkings of the malicious attachment describing it as a trojan.
The 5 W's	Capture the 5 W's of an incident. • Who caused the incident? A threat actor that could be linked with an APT • What happened? HR employee received a phishing email with a malicious attachment. The attachment which was a Trojan was executed on the employee's machine. • When did the incident occur? Wednesday, July 20, 2022 09:30:14 AM • Where did the incident happen? HR employee machine. • Why did the incident happen? Lack of security awareness
Additional notes	

Date: Friday, March 27, 2026 3:07 PM	Entry: 3
Description	Investigating an alert on suricata.
Tool(s) used	Suricata an open source IDS, IPS , and a network analyzer.
The 5 W's	Capture the 5 W's of an incident. • Who caused the incident? A local machine with the IP 172.21.224.2 and port 49652 • What happened? A local machine triggered the custom rule GET on wire. • When did the incident occur? 11/23/2022-12:38:34.624866 • Where did the incident happen? On the local organization network. • Why did the incident happen? The incident occurred when the machine operator visited a website, pulling a web page in the process.
Additional notes	The custom suricata rule was designed to trigger on every GET pull done from local machines. The rule was the following <code>alert http \$HOME_NET any -> \$EXTERNAL_NET any (msg:"GET on wire"; flow:established,to_server; content:"GET"; http_method; sid:12345; rev:3;).</code> The command jq was used to parse through eve.json files. <pre>jq -c "[.timestamp,.flow_id,.alert.signature,.proto,.dest_ip]" /var/log/suricata/eve.json</pre> <pre>jq "select(.flow_id==X)" /var/log/suricata/eve.json</pre>

Date: Friday, March 27, 2026 3:27 PM	Entry: 4
Description	I need to use tcpdump to capture and analyze live network traffic from a Linux virtual machine.
Tool(s) used	Tcpdump an open source tool used to capture and analyzed captured packets.
The 5 W's	Capture the 5 W's of an incident. • Who caused the incident? I launched the tcpdump programme. • What happened? Packets have been captured using tcpdump and analyzed afterwards using the same afromentioned tool. • When did the incident occur? The app was launched on Monday, March 23, 2026 8 AM • Where did the incident happen? On the virutal machine hosted by google on the google skills website. • Why did the incident happen? Curiosity to learn
Additional notes	To identify interfaces with tcpdump <pre>sudo tcpdump -D</pre> <pre>sudo tcpdump -i eth0 -v -c5</pre> This command will run tcpdump with the following options: -i eth0: Capture data specifically from the eth0 interface. -v: Display detailed packet data. -c5: Capture 5 packets of data. <pre>sudo tcpdump -i eth0 -nn -c9 port 80 -w capture.pcap &</pre> This command will run tcpdump in the background with the following options:

-i eth0: Capture data from the eth0 interface.

-nn: Do not attempt to resolve IP addresses or ports to names. This is best practice from a security perspective, as the lookup data may not be valid. It also prevents malicious actors from being alerted to an investigation.

-c9: Capture 9 packets of data and then exit.

port 80: Filter only port 80 traffic. This is the default HTTP port.

-w capture.pcap: Save the captured data to the named file.

&: This is an instruction to the Bash shell to run the command in the background.

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sudo tcpdump -nn -r capture.pcap -X
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This command will run tcpdump with the following options:

-nn: Disable port and protocol name lookup.

-r: Read capture data from the named file.

-X: Display the hexadecimal and ASCII output format packet data. Security analysts can analyze hexadecimal and ASCII output to detect patterns or anomalies during malware analysis or forensic analysis.

Reflections/Notes:

Using this journal i managed to document what i learned in a standarized manner. This has shown me how powerful documentation can be. During my journey studying for this certificate i was able to go back the my incident response journal and recall the way to use commands when i needed them. Documentation can also boost retention i noticed that i retain much more information if i document it after studying it. Documentaion helped me practice my english technical writing skills as well. Overall boosting clarity, satandarization, and Transparency I personnaly think documentaion is a powerful tool for students and projects a like.

Date: Friday, March 27, 2026 4:38 PM	Entry: 5
Description	Investigated a potential brute force attack on a mail server by querying SIEM logs for failed root login attempts.
Tool(s) used	Wazuh An open-source security monitoring and SIEM platform used to search and analyze log data.
The 5 W's	Capture the 5 W's of an incident. • Who caused the incident? An unknown threat actor attempting to guess administrative credentials. • What happened? Over 300 failed SSH login attempts were made targeting the 'root' account on the mail server. • When did the incident occur? N/A broken VM. • Where did the incident happen? Buttercup Games' mail server identified in logs as the host: mailsv. • Why did the incident happen? No rate limiting set in place.
Additional notes	* acts as a wildcard to pull all data. host.keyword: mailsv filters data to only show logs originating from the mail server. query used: host.keyword: mailsv AND (fail* OR failed) AND root

Date: Record the date of the journal entry.	Entry: Record the journal entry number.
Description	Provide a brief description about the journal entry.
Tool(s) used	List any cybersecurity tools that were used.
The 5 W's	Capture the 5 W's of an incident. • Who caused the incident? • What happened? • When did the incident occur? • Where did the incident happen? • Why did the incident happen?
Additional notes	Include any additional thoughts, questions, or findings.